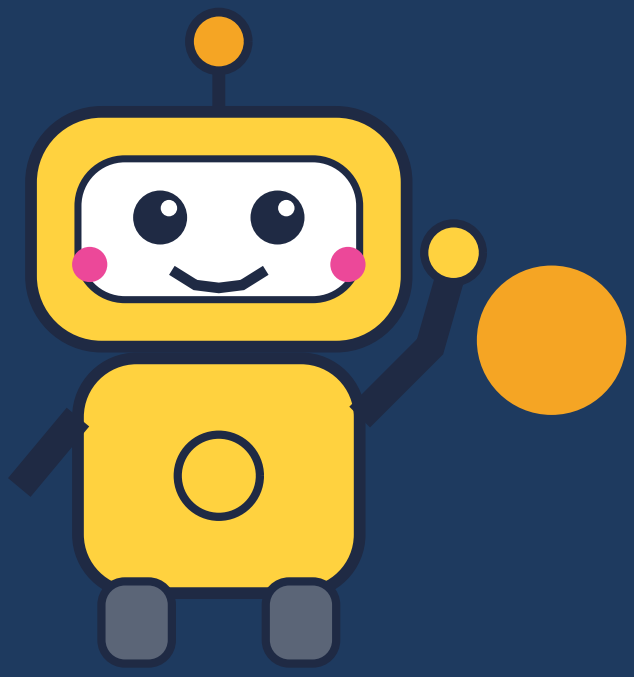
SAMPLE PAPERS

Think Like a Coder

Sample examination papers, Class 7

Half-yearly and annual papers
50 marks, 2 hours, with blueprints and marking schemes
Practical stations and project assessment



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 7 · Photocopiable for school use

SAMPLE EDITION · DRAFT

© 2026 Bullseye Knowledge System Limited. All rights reserved.
CodeSchool is a working name. Not an official CBSE or NCERT publication.

What's inside

About these papers 3

Half-yearly examination (question paper) 4

Annual examination (question paper) 7

Blueprints 10

Marking schemes 12

Practical and project assessment 13

© 2026 Bullseye Knowledge System Limited

About these papers

These are two sample examination papers for Class 7: a **half-yearly** paper on the term-1 chapters and an **annual** paper on the whole year. They go with *Think Like a Coder*, Class 7, and with the chapter assessments. All questions are new.

How they follow CBSE

- CBSE does not hold board examinations for Classes 3-8 and has not published a question-paper pattern for CT&AI.; Schools set their own papers. These papers follow what CBSE's CT&AI; handbooks describe: **competency-based** written tests with CT puzzles, with no 'write a program' questions and no computer needed.
- The question types are the ones CBSE uses in its competency-based papers: multiple choice, assertion-reason, short and long answers, and a case-based question.
- Questions are spread across four thinking levels (remember, understand, apply, analyse and create), with most marks for applying and analysing. The blueprints show the split.
- Chapters follow the order of CBSE's 2026-27 CT&AI; handbooks and the NCERT maths chapters, including the four AI chapters.
- The handbooks also assess Classes 6-8 through practical examinations, project presentations and reflective journals. The last section gives unplugged practical stations, a rubric and one way to combine the marks.

Paper design

Section	Question type	Questions	Marks
A	Multiple choice and assertion-reason	10	10 x 1 = 10
B	Very short answer	5	5 x 2 = 10
C	Short answer	5	5 x 3 = 15
D	Long answer	2	2 x 5 = 10
E	Case-based question	1	1 x 5 = 5
	Total · 2 hours	23	50

Using the papers

- Use the half-yearly paper in September and the annual paper in February-March, or as mock tests before the school's own exams.
- Print one paper per student; students answer on separate answer sheets.
- Read questions aloud to students who find reading hard: the paper tests thinking, not reading.
- Mark with the marking scheme; accept any correct method, and give part marks for correct steps.
- Use the blueprints to set your own papers in the same pattern.

© 2026 Bullseye Knowledge System Limited

SAMPLE PAPER · HALF-YEARLY EXAMINATION

Class 7 · Computational Thinking and Artificial Intelligence

Syllabus: chapters 1-4 and AI 1-2 (term 1)

Time allowed: 2 hours **Maximum marks:** 50

Name: _____ **Section:** _____ **Roll no.:** _____

General instructions

1. This paper has 23 questions in 5 sections. All questions are compulsory.
2. Section A: multiple choice and assertion-reason, 10 questions of 1 mark each; Section B: very short answer, 5 questions of 2 marks each; Section C: short answer, 5 questions of 3 marks each; Section D: long answer, 2 questions of 5 marks each; Section E: case-based question, 1 question of 5 marks each.
3. Marks for each question are shown on the right.
4. Write your answers on the separate answer sheet. Draw tables and diagrams neatly where asked.
5. No calculator or computer is needed. Show your working: marks are given for correct steps.

SECTION A · Multiple choice and assertion-reason — 10 x 1 = 10 marks

1. 10 lakh = ... [1]

a) 1 million **b)** 10 million **c)** 1 billion

2. 6,48,512 rounded to the nearest lakh is ... [1]

a) 6,00,000 **b)** 6,50,000 **c)** 7,00,000

3. 12 - 8 ÷ 4 = ... [1]

a) 1 **b)** 10 **c)** 2

4. The binary number 1101 is ... in decimal. [1]

a) 11 **b)** 13 **c)** 1101

5. In 0.06, the 6 is in the ... place. [1]

a) tenths **b)** hundredths **c)** thousandths

6. If a = 3, then 2a + 5 = ... [1]

a) 11 **b)** 28 **c)** 10

7. Grouping customers with similar shopping habits, with no labels, is ... [1]

a) classification **b)** regression **c)** clustering

8. AI that picks the next practice question from a student's mistakes is used in ... [1]

a) education **b)** transport **c)** farming

© 2026 Bullseye Knowledge System Limited

9. **Assertion (A):** In 5 + 3 x 2, the multiplication is done first. [1]
Reason (R): Multiplication and division are done before addition and subtraction.

a) Both A and R are true, and R explains A.

b) Both A and R are true, but R does not explain A.

c) A is true, but R is false.

d) A is false, but R is true.

10. **Assertion (A):** A model should be tested on the same data it was trained on. [1]
Reason (R): Test data shows how well the model works on data it has not seen.

a) Both A and R are true, and R explains A.

b) Both A and R are true, but R does not explain A.

c) A is true, but R is false.

d) A is false, but R is true.

SECTION B · Very short answer — 5 x 2 = 10 marks

11. Write 25,000,000 with Indian commas and in words. [2]

12. Write one expression for 'add 4 and 6, then multiply by 3', and find its value. [2]

13. Write 0.75 as a fraction in simplest form, and 3/8 as a decimal. [2]

14. The perimeter of a rectangle is 2(l + b). Find it when l = 7 cm and b = 4.5 cm. [2]

15. Give one benefit and one risk of using AI in healthcare. [2]

SECTION C · Short answer — 5 x 3 = 15 marks

16. Draw the expression tree for 3 x (8 - 2) + 10 ÷ 5 and work out its value. [3]

17. Write 5, 6, 7 and 8 as 4-bit binary numbers. How many bits are needed to write every number up to 100? [3]

18. Tables in a row seat 4n + 2 people, where n is the number of tables. How many people do 12 tables seat? How many tables are needed for 38 people? [3]

19. Classification, regression or clustering? (a) Predicting a house's price from its size. (b) Sorting X-rays into 'healthy' and 'not healthy'. (c) Grouping news stories by topic when there are no labels. [3]

20. Estimate 4,87,215 + 2,13,980 by rounding each number to the nearest thousand. Then find the exact answer and the difference. [3]

© 2026 Bullseye Knowledge System Limited

SECTION D · Long answer — 2 x 5 = 10 marks

21. A dataset has 500 photos of leaves, each labelled 'healthy' or 'diseased'.

(a) Which AI domain is this? [1]

(b) Is the task classification, regression or clustering? [1]

(c) How would you split the 500 photos? [1]

(d) The model is right 96% of the time on training photos but only 70% on test photos. What has gone wrong? [1]

(e) Give one way this model could help farmers. [1]

22. Hexagons in a row are made from sticks: 1 hexagon uses 6 sticks, 2 use 11 and 3 use 16.

(a) Write a formula for the sticks needed for n hexagons. [2]

(b) How many sticks do 40 hexagons need? [1]

(c) How many hexagons can 61 sticks make? [1]

(d) Write the formula as one line of a program. [1]

SECTION E · Case-based question — 1 x 5 = 5 marks

23. A route app compares two routes. Route A is 14 km and live traffic moves at 28 km/h. Route B is 10 km and live traffic moves at 15 km/h.

(a) How long does each route take? [2]

(b) Which route does the app choose? Why? [1]

(c) What live data does the app use? [1]

(d) Give one privacy concern about route apps. [1]

***** End of the question paper *****

© 2026 Bullseye Knowledge System Limited

SAMPLE PAPER · ANNUAL EXAMINATION

Class 7 · Computational Thinking and Artificial Intelligence

Syllabus: all chapters, mostly chapters 5-8 and AI 3-4 (term 2)

Time allowed: 2 hours **Maximum marks:** 50

Name: _____ **Section:** _____ **Roll no.:** _____

General instructions

1. This paper has 23 questions in 5 sections. All questions are compulsory.
2. Section A: multiple choice and assertion-reason, 10 questions of 1 mark each; Section B: very short answer, 5 questions of 2 marks each; Section C: short answer, 5 questions of 3 marks each; Section D: long answer, 2 questions of 5 marks each; Section E: case-based question, 1 question of 5 marks each.
3. Marks for each question are shown on the right.
4. Write your answers on the separate answer sheet. Draw tables and diagrams neatly where asked.
5. No calculator or computer is needed. Show your working: marks are given for correct steps.

SECTION A · Multiple choice and assertion-reason — 10 x 1 = 10 marks

1. Two lines cross and one angle is 110°. The angle next to it on the straight line is ... [1]

a) 70° **b)** 110° **c)** 250°

2. With 5 lights, each ON or OFF, how many cases are there? [1]

a) 10 **b)** 25 **c)** 32

3. A triangle with all three sides different is ... [1]

a) scalene **b)** isosceles **c)** equilateral

4. 1/2 x 4/5 = ... [1]

a) 2/5 **b)** 5/10 **c)** 5/7

5. Parts of a whole are best shown with a ... [1]

a) pie chart **b)** line graph

6. Data that repeats unfair decisions from the past shows ... [1]

a) data bias **b)** historical bias

c) measurement bias

7. The binary number 10000 is ... in decimal. [1]

a) 16 **b)** 32 **c)** 10

8. In a pie chart for 36 students, the slice for 9 students is ... [1]

a) 36° **b)** 90° **c)** 9°

© 2026 Bullseye Knowledge System Limited

9. **Assertion (A):** 3 cm, 4 cm and 9 cm can make a triangle. [1]
Reason (R): In a triangle, any two sides added together are longer than the third side.

a) Both A and R are true, and R explains A.

b) Both A and R are true, but R does not explain A.

c) A is true, but R is false.

d) A is false, but R is true.

10. **Assertion (A):** Corresponding angles made by a transversal on two parallel lines are equal. [1]
Reason (R): Parallel lines never meet.

a) Both A and R are true, and R explains A.

b) Both A and R are true, but R does not explain A.

c) A is true, but R is false.

d) A is false, but R is true.

SECTION B · Very short answer — 5 x 2 = 10 marks

11. A transversal crosses two parallel lines, and one angle is 72°. Find the co-interior angle that pairs with it. [2]

12. Lights R, G and B. Clue 1: exactly one light is OFF. Clue 2: Red is OFF. Which lights are ON? [2]

13. An isosceles triangle has a top angle of 40°. Find the two equal base angles. [2]

14. Half of a number, plus 7, is 19. Work backwards to find the number. [2]

15. Name two kinds of bias in AI, with an example of each. [2]

SECTION C · Short answer — 5 x 3 = 15 marks

16. Explain why the sum of any three consecutive numbers is a multiple of 3. [3]

17. Write a function classify(a, b, c) that names a triangle by its sides. Test it on (5, 5, 5), (5, 5, 8) and (4, 6, 7). [3]

18. A tank is 3/4 full. 1/3 of the water in it is used, and 30 litres are left. What is the full capacity of the tank? [3]

19. Of 120 students, 30 like cricket, 45 football, 15 hockey and 30 badminton. Find the angle of each slice in a pie chart, and check the total. [3]

20. A turtle starts with heading 0°. Program: forward 40, turn right 50°, forward 30, turn right 130°, forward 40. Which two lines are parallel? Explain using headings. [3]

© 2026 Bullseye Knowledge System Limited

SECTION D · Long answer — 2 x 5 = 10 marks

21. A bank trains a loan-approval model only on past decisions from one city branch.

(a) What kind of bias could this cause? [1]

(b) How would you check whether the model is fair? [1]

(c) How would you rebalance the data? [1]

(d) What should humans do when the model is used? [1]

(e) Give one way to make the model's decisions transparent. [1]

22. Three lights R, G and B are each ON or OFF.

(a) List all 8 cases. [2]

(b) Clue 1: at least two lights are ON. Which cases are left? [1]

(c) Clue 2: Blue is OFF. Which case is left? [1]

(d) How many cases would there be with 6 lights? [1]

SECTION E · Case-based question — 1 x 5 = 5 marks

23. The temperature at noon on six days was 31°C, 33°C, 32°C, 35°C, 38°C and 35°C.

(a) Which chart would you use? Why? [1]

(b) What is the highest temperature, and the range? [1]

(c) What is the average temperature? [1]

(d) Describe the trend. [1]

(e) A chart's temperature axis starts at 30°C. Is that honest? Explain. [1]

***** End of the question paper *****

© 2026 Bullseye Knowledge System Limited

Blueprints

Blueprint: Half-yearly examination

Syllabus: chapters 1-4 and AI 1-2 (term 1). Marks by chapter and section.

Chapter	A	B	C	D	E	Marks
1. Large Numbers Around Us	2	2	3			7
2. Arithmetic Expressions	2	2	3			7
3. A Peek Beyond the Point	2	2	3			7
4. Expressions using Letter-Numbers	1	2	3	5		11
AI 1. AI Domains and Applications	2		3	5		10
AI 2. AI in Industries	1	2			5	8
Total	10	10	15	10	5	50

Marks by thinking level

Each question is counted at its main thinking level.

Level	What the student does	Marks	Share
Remember	recall a fact or term	2	4%
Understand	explain or interpret an idea	8	16%
Apply	use an idea or rule in a new problem	26	52%
Analyse and create	trace, debug, compare, design or justify	14	28%

© 2026 Bullseye Knowledge System Limited

Blueprint: Annual examination

Syllabus: all chapters, mostly chapters 5-8 and AI 3-4 (term 2). Marks by chapter and section.

Chapter	A	B	C	D	E	Marks
3. A Peek Beyond the Point	1					1
5. Parallel and Intersecting Lines	2	2	3			7
6. Number Play	1	2	3	5		11
7. A Tale of Three Intersecting Lines	2	2	3			7
8. Working with Fractions	1	2	3			6
AI 3. Data Visualisation and Analysis	2		3		5	10
AI 4. Ethics and AI Bias Awareness	1	2		5		8
Total	10	10	15	10	5	50

Marks by thinking level

Each question is counted at its main thinking level.

Level	What the student does	Marks	Share
Remember	recall a fact or term	2	4%
Understand	explain or interpret an idea	4	8%
Apply	use an idea or rule in a new problem	29	58%
Analyse and create	trace, debug, compare, design or justify	15	30%

© 2026 Bullseye Knowledge System Limited

Marking schemes

Accept any correct method and any sensible wording. Give part marks for correct steps, even when the final answer has a slip.

Marking scheme: Half-yearly examination

1. a) 1 million
2. a) 6,00,000
3. b) 10
4. b) 13
5. b) hundredths
6. a) 11
7. c) clustering
8. a) education
9. a) Both A and R are true, and R explains A.
10. d) A is false, but R is true.
11. 2,50,00,000; two crore fifty lakh (1 mark for each)
12. (4 + 6) x 3 = 30 (1 mark for the brackets, 1 mark for 30)
13. 3/4; 0.375 (1 mark for each)
14. 2(7 + 4.5) = 23 cm (1 mark for substituting, 1 mark for 23 cm)
15. For example: benefit, it spots illness early in X-rays; risk, it can be wrong, so a doctor must check (1 mark for each)
16. Top: +. Left branch: x joining 3 and (- joining 8 and 2). Right branch: ÷ joining 10 and 5. Value: 18 + 2 = 20. (1 mark for + at the top, 1 mark for both branches, 1 mark for 20)
17. 0101, 0110, 0111, 1000. 7 bits (6 bits reach only 63; 7 bits reach 127). (2 marks for the four numbers (1 mark for two), 1 mark for 7 bits)
18. 50 people; 9 tables (4 x 9 + 2 = 38) (1 mark for 50, 2 marks for 9 tables (1 mark for the method))
19. (a) regression (b) classification (c) clustering (1 mark for each)
20. 4,87,000 + 2,14,000 = 7,01,000; exact 7,01,195; difference 195 (1 mark for each)
21. (a) Computer Vision (b) classification (c) for example 350 training, 75 validation, 75 test (d) It has learnt the training photos too closely; it needs more varied photos and must be judged on test data (e) for example, farmers can spot disease early and save their crop (1 mark for each part)
22. (a) 5n + 1 (b) 201 (c) 12 hexagons (d) set sticks to 5 x n + 1 ((a) 2 marks (b) 1 mark (c) 1 mark (d) 1 mark)
23. (a) A: 30 minutes; B: 40 minutes (b) Route A, because it takes less time even though it is longer (c) how fast phones on each road are moving (d) for example, the app tracks where people go, so location data must be kept private ((a) 2 marks (b) 1 mark (c) 1 mark (d) 1 mark)

Marking scheme: Annual examination

1. a) 70°
2. c) 32
3. a) scalene
4. a) 2/5
5. a) pie chart
6. b) historical bias
7. a) 16
8. b) 90°
9. d) A is false, but R is true.
10. b) Both A and R are true, but R does not explain A.
11. 108° (1 mark for 180° - 72°, 1 mark for 108°)
12. Green and Blue (2 marks)
13. 70° each (1 mark for 140° ÷ 2, 1 mark for 70°)
14. 19 - 7 = 12; 12 x 2 = 24 (1 mark for each step)
15. For example: data bias, a face detector trained mostly on adults; historical bias, hiring data that repeats past unfairness (1 mark for each)
16. n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1), which is a multiple of 3. (1 mark for the three numbers, 1 mark for 3n + 3, 1 mark for 3(n + 1))
17. If a = b and b = c: equilateral; else if a = b or b = c or a = c: isosceles; else: scalene. Results: equilateral, isosceles, scalene. (1 mark for the function, 2 marks for all three results (1 mark for two))
18. 30 L is 2/3 of the water, so the water was 45 L. 45 L is 3/4 of the tank, so the tank holds 60 L. (1 mark for 45 L, 1 mark for the method, 1 mark for 60 L)
19. 90°, 135°, 45°, 90°; total 360° (2 marks for the angles (1 mark for two), 1 mark for the check)
20. The headings are 0°, 50° and 180°. The first and third lines are parallel: 180° points the opposite way along the same direction. (1 mark for the headings, 1 mark for the answer, 1 mark for the reason)
21. (a) data bias, and possibly historical bias (b) compare approval rates and accuracy for different groups and places (c) add records from other branches and groups (d) a person reviews important decisions and takes responsibility (e) tell people the main reasons for a decision (1 mark for each part)
22. (a) OOO, OOU, OUO, OUU, UOO, UOU, UUO, UUU (U = ON, O = OFF) (b) R and G; R and B; G and B; all three (4 cases) (c) only Red and Green ON (d) 64 ((a) 2 marks, 1 mark for six or seven cases (b) 1 mark (c) 1 mark (d) 1 mark)
23. (a) a line graph, because it shows change over time (b) 38°C; range 7°C (c) 34°C (d) mostly rising, with a dip on day 3 and on day 6 (e) It can be honest if the axis is clearly marked, but it makes small changes look large; starting at 0 or marking the break is fairer (1 mark for each part)

© 2026 Bullseye Knowledge System Limited

Practical and project assessment

CBSE's Classes 6-8 handbooks assess CT&AI; through practical examinations, project presentations and reflective journals as well as written tests. CBSE has not fixed how the marks are split, so the school decides. One way to make a 100-mark annual result:

Component	Marks	Evidence
Written examination (this paper)	50	Annual sample paper
Practical examination (unplugged)	20	Two stations from the list below, 10 marks each
Projects (two 20-hour projects)	20	Project rubrics in the workbook; presentation and data sheets
Observation journal and reflection	10	Teacher Observation Journal levels from the teacher book

Practical examination stations

Each station takes about 15 minutes. Students work alone or in pairs with paper, cards or the classroom kit; the examiner watches, asks the two questions shown, and marks with the rubric. No computer is needed.

Station	Task	Examiner asks
1. Binary cards (chapter 3)	Use five dot cards (16, 8, 4, 2, 1) to show three numbers the examiner chooses, then read two binary numbers back.	Why does each card double? How many numbers can five cards show?
2. Expression trees (chapter 2)	Build an expression tree from cards for an expression, then evaluate it from the bottom up.	Which operation is done last? How do brackets change the tree?
3. Angle tracer (chapter 5)	Trace a turtle program's heading after each turn and mark which lines are parallel.	How do headings show parallel lines?
4. Clue filter (chapter 6)	Lay out all 8 light cards and remove cards clue by clue until one is left.	Why start with every case? What does each clue remove?
5. AI task sort (AI 1)	Sort twelve problem cards into classification, regression and clustering.	What does each kind of task give as its answer?
6. Survey to pie chart (AI 3)	Turn a small survey table into pie-chart angles and draw the chart.	How do you find each angle? When is a bar chart better?
7. The lopsided deck (AI 4)	Find the bias in a training deck of cards and rebalance it.	Which group is missing? How would you test the fix?

Rubric for each station (10 marks)

Criterion	Marks	Full marks when the student ...
Understands the task	2	restates the goal and what a correct result looks like
Method	4	follows or designs clear, ordered steps; uses the right CT or AI idea
Result and checking	2	gets a correct result and checks or tests it
Explaining	2	answers the examiner's questions with reasons, using the right words

© 2026 Bullseye Knowledge System Limited